

Step 1 (Truth Table for Output a):-

BCD (A B C)	Decimal	Output a (Segment a)
0000	0	1
0001	1	0
0010	2	1
0011	3	1
0100	4	0
0101	5	1
0110	6	1
0111	7	0
1000	8	1
1001	9	1

Step 2 (Formulate the Binary SOP for Output a):-

From the truth table, we can derive the **Sum of Products (SOP)** expression.

- 000000000000 (Decimal 0)
- 001000100010 (Decimal 2)
- 001100110011 (Decimal 3)
- 010101010101 (Decimal 5)
- 011001100110 (Decimal 6)
- 100010001000 (Decimal 8)
- 100110011001 (Decimal 9)

The binary SOP expression for 'a' can be written as:-

$$a = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}\overline{D}$$

Step 3 (Draw the K-map for Output a):-

		CD				
		00	01	11	10	
AB	00	1	0	0	1	0 (0000, 0001, 0010, 0011)
	01	1	1	0	1	5 (0100, 0101, 0110, 0111)
	11	1	1	0	1	8 (1000, 1001, 1010, 1011)
	10	0	0	0	0	4 (1100, 1101, 1110, 1111)

Step 3 (Simplify the Binary Expression Using the K-map):-

Grouping the 1s in the K-map:

1. Group 1 (four 1s):

- Covers: (0000, 0010, 0011, 0100) → Output a = $\overline{A}\overline{B}$

2. Group 2 (two 1s):

- Covers: (0101, 0110) → Output a = $\overline{A}BC$

3. Group 3 (two 1s):

- Covers: (1000, 1001) → Output a = $A\overline{B}$

Final Expression:-

$$a = \overline{A}\overline{B} + \overline{A}BC + A\overline{B}$$